Letters in Applied Microbiology

Letters in Applied Microbiology ISSN 0266-8254

ORIGINAL ARTICLE

Virulence associated gene profiling and antimicrobial resistance pattern of *Streptococcus suis* isolated from clinically healthy pigs from North East India

S. Rajkhowa[1] and J.B. Rajesh[2]

1 Animal Health Laboratory, ICAR-National Research Centre on Pig, Rani, Guwahati, Assam, India
2 College of Veterinary Sciences & Animal Husbandry, Central Agricultural University, Selesih, Mizoram, India

Significance and Impact of the Study: This study provides valuable information on the virulence-associated genes and antimicrobial resistance (AMR) of *Streptococcus suis* isolates from clinically healthy pigs, which can guide for judicious use of antimicrobial drugs and will also be helpful for monitoring of AMR in *S. suis*.

Keywords
antimicrobial resistance, clinically healthy pigs, India, *Streptococcus suis*, virulence- associated genes.

Correspondence
Swaraj Rajkhowa, Animal Health Laboratory, ICAR-National Research Centre on Pig, Rani, Guwahati, Assam, India.
Email: swaraj.rajkhowa@gmail.com

2021/0475: received 5 March 2021, revised 24 June 2021 and accepted 25 June 2021

doi:10.1111/lam.13531

Abstract

This study revealed the prevalence of *Streptococcus suis* in 20·39% clinically healthy pigs from North East India. All these isolates were screened for the presence of virulence- associated genes such as suilysin (*sly*), muramidase released protein (*mrp*), extracellular protein factor (*epf*) and arginine deiminase (*arcA*). Of these 62 isolates, 29 isolates carried *mrp* gene, 17 isolates carried *sly* gene, 57 isolates carried *arcA* gene, whereas all isolates were negative for *epf* gene. The most prevalent genotype was mrp^{-} sly^{-} epf^{-} $arcA^{+}$ (45·16%) followed by genotypes mrp^{+} sly^{-} epf^{-} $arcA^{+}$ (27·41%), mrp^{+} sly^{+} epf^{-} $arcA^{+}$ (19·35%) and mrp^{-} sly^{+} epf^{-} $arcA^{-}$ (8·06%). High frequency of resistance was observed for antimicrobials such as tetracycline (93·54%), clindamycin (91·93%), co-trimoxazole (88·70%) and erythromycin (85·48%). Antimicrobial resistance patterns of the *S. suis* isolates revealed 16 resistance groups (R1 to R16), where 93·54% isolates showed multi-drug resistance (≥3 antimicrobial agents). It has also been observed that 57 (91·93%) isolates were resistant to at least four antimicrobials. The most predominant resistance pattern observed was CD-COT-E-TE, which accounted for 38·70% of the isolates. The occurrence of relatively high levels of resistance of *S. suis* to some antimicrobials (e.g., macrolides, tetracyclines, and sulphonamides) as observed in this study may represent a human health concern. In addition, a relatively higher percentage of *S. suis* isolated from clinically healthy pigs indicates a carrier status with risk of dissemination to other pigs in the herd as well as to humans.

Introduction

Streptococcus suis is an encapsulated Gram-positive coccus with approximately 30 currently recognized serotypes that are defined on the different antigenicity of capsular polysaccharides (Lacouture *et al.* 2020). *Streptococcus suis* can be found in the upper respiratory tract of healthy pigs particularly in piglets, which can act as carriers of the disease (MacInnes and Desrosiers 1999). *Streptococcus suis* is a zoonotic pathogen that causes serious diseases, including meningitis, arthritis, septicemia, and endocarditis in pigs and humans and is responsible for severe economic losses in the swine industry worldwide (Goyette-Desjardins *et al.* 2014). Healthy carrier pigs are considered as a potential health hazard of workers in the pig and pork industry. Although systematic research of the

S. suis carrier state in pigs was performed in many countries, to the best our knowledge such studies from India are very scanty. Healthy carrier pigs are important not only regarding the spread of *S. suis* in herds but also as source of infection for humans (Lun *et al.* 2007). Human infection usually is caused by direct contact with carrier pigs or consumption of raw pork contaminated with *S. suis*. In Western countries, *S. suis* infections in humans have generally been restricted to workers in close contact with pigs or swine by-products. However, in East and Southeast Asia, *S. suis* represents a significant public health concern (Goyette-Desjardins *et al.* 2014), as this bacterium was identified as the first, second, and third causes of adult meningitis in Viet Nam, Thailand, and Hong Kong, respectively (Suankratay *et al.* 2004; Ip *et al.* 2007; Mai *et al.* 2008).

Muramidase-released protein (MRP), extracellular factor, and suilysin (SLY), encoded by the genes *mrp*, *epf*, and *sly*, respectively, are considered virulence markers and are used to predict the virulence of a given *S. suis* strain. It was also pointed out that the enzyme arginine deiminase in *S. suis* (encoded by the *arcA* gene of the arginine deiminase operon), seems to play a role in survival of the bacterium under stress conditions (Gruening *et al.* 2006).

Antimicrobial drugs have long been applied as therapeutic and/or prophylactic medication for *S. suis* infection in pigs and humans. However, the imprudent use of antimicrobial drugs results in serious antimicrobial resistance (AMR) that has become a global problem in recent years. Increasing cases of AMR in *S. suis* isolated from pigs and humans were observed from various countries in Asia, Europe, and North America (Zhang *et al.* 2008; Varela *et al.* 2013). Such cases increase the risks of therapeutic failure in animals and humans. Among antimicrobials examined, a high rate of resistance to lincosamides, macrolides, sulphonamides, and tetracyclines (TETs) has been frequently reported (Varela *et al.* 2013).

Monitoring AMR profiles and the virulence-associated markers of *S. suis* in certain regions is necessary for optimizing effective antimicrobial treatments and following up the emergence of bacterial drug resistance. However, the virulence markers and AMR information of *S. suis* circulating in clinically healthy pigs from North East India, a major pig rearing area of India, has not been reported yet. In the present study, *S. suis* isolates recovered from the nasal cavity of clinically healthy pigs were assessed through virulence- associated markers detection and antimicrobial susceptibility testing.

Results and discussion

In this study, we recorded the prevalence of *S. suis* in 20·39% (62/304) clinically healthy pigs. Although scanty information is available from India, several studies from different parts of the globe have examined the prevalence of *S. suis* in clinically healthy pigs which ranged from 2·8 to 37% (Xiong *et al.* 2007; Meekhanon *et al.* 2017). The observed differences in this study could be due to differences in husbandry practices and prevailing climatic conditions, which may account for the varied prevalence of *S. suis* from one geographical region to another (Votsch *et al.* 2018).

BLAST analysis of the GDH sequence in this study demonstrated that the sequence had query cover of 100% and an identity score of 94·65–95·16% with the sequence of human S. *suis* isolates (Serotype 24) from Thailand (GenBank accession no. CP076517.1 and CP068708.1) and with the sequence of *S. suis* isolates (Serotype 2) from both pigs and human from USA, England, and Canada (GenBank accession no. EF198475.1).

All (62) the isolates of *S. suis* were screened for the presence of virulence- associated genes like *mrp*, *epf*, *sly*, and *arcA*. The distribution of virulence-associated genes is shown in Table 1.

Of these 62 isolates, 29 isolates carried *mrp* gene (Fig. 1), 17 isolates carried *sly* gene (Fig. 2), 57 isolates carried *arcA* gene (Fig. 3) whereas all isolates were negative for *epf* gene. The *epf*, *mrp*, and *sly* genes have been shown to be significant virulence markers for *S. suis* that allow to differentiate virulent (presence) from less-virulent (absence) strains in Europe and Asia (Schultsz *et al.* 2012).

The presence of other additional genes, such as *arcA* has also been reported to be associated with *S. suis* virulence (Winterhoff *et al.* 2002). The present findings corroborated with the findings of several authors (Maneerat *et al.* 2013; Wang *et al.* 2013) who also observed that *S. suis* isolates from clinically healthy pigs carried these virulence genes. Although several workers (Rui *et al.* 2012; Espinosa *et al.* 2013) reported detection of *epf* gene in *S. suis* isolates from pigs, all the isolates in this study were negative for this gene, which is consistent with the findings of Zheng *et al.* (2014) who also found that most of the *S. suis* isolates obtained from China were *epf* negative. The present finding is almost similar with the findings of

Table 1 Virulence genotypes of 62 *Streptococcus suis* isolates from clinically healthy pigs

Genotypes	Number (%) of *S. suis*
mrp^+ sly^+ epf^- $arcA^+$	12 (19·35)
mrp^+ sly^- epf^- $arcA^+$	17 (27·41)
mrp^- sly^+ epf^- $arcA^-$	5 (8·06)
mrp^- sly^- epf^- $arcA^+$	28 (45·16)

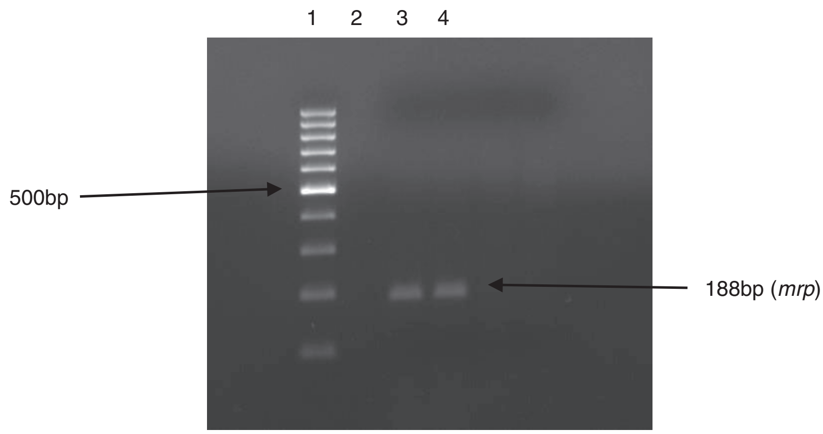

Figure 1 Detection of *mrp* gene of *Streptococcus suis* from pig using PCR, Lane 1: molecular marker (100 bp DNA ladder), Lane 2: negative control, Lane 3: positive control, Lane 4: test organism.

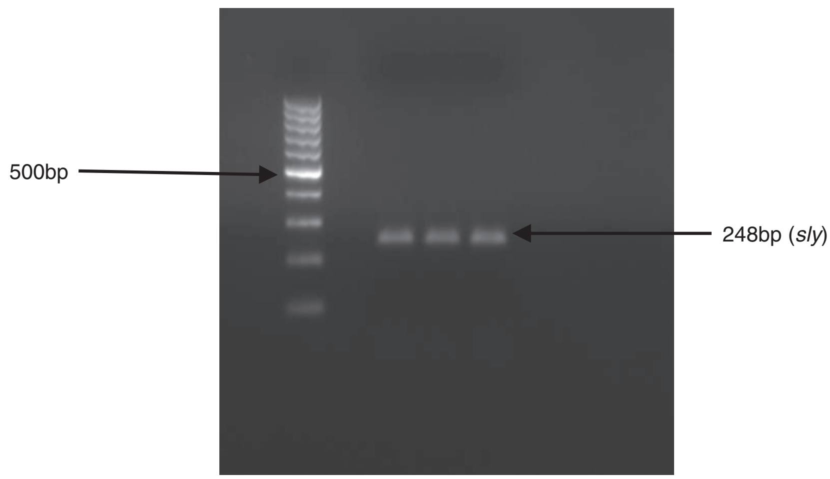

Figure 2 Detection of *sly* gene of *Streptococcus suis* from pig using PCR, Lane 1: molecular marker (100 bp DNA ladder), Lane 2: negative control, Lane 3: positive control, lane 4–5: test organism.

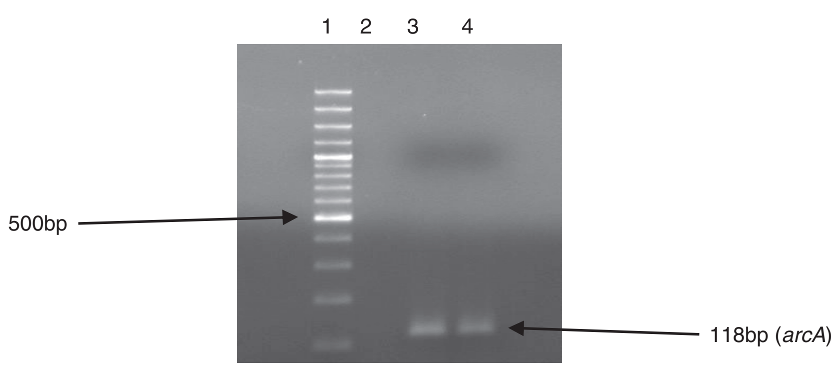

Figure 3 Detection of *arcA* gene of *Streptococcus suis* from pig using PCR, Lane 1: molecular marker (100 bp DNA plus ladder), Lane 2: negative control, Lane 3: positive control, Lane 4: test organism.

Maneerat *et al.* (2013) who observed that all *S. suis* isolates from asymptomatic pigs were positive for *arcA* gene.

Four different virulence genotypes were obtained in this study with highest frequency of genotype mrp^{-} sly^{-} epf^{-} $arcA^{+}$ (45·16%) followed by genotypes mrp^{+} sly^{-} epf^{-} $arcA^{+}$ (27·41%), mrp^{+} sly^{+} epf^{-} $arcA^{+}$ (19·35%), and mrp^{-} sly^{+} epf^{-} $arcA^{-}$ (8·06%). It was also reported that based on the genes of *mrp*, *sly*, and *epf*, the most commonly observed genotype of *S. suis* was $mrp^{-}sly^{-}epf^{-}$ (Wang *et al.* 2013). Similarly, Meekhanon *et al.* (2017) reported that the most common *S. suis* genotype circulating among asymptomatic pigs in Thailand was $mrp^{-}sly^{-}epf^{-}$.

In this study, high frequency of resistance was observed for antimicrobials such as TET (93·54%), clindamycin (91·93%), co- trimoxazole (88·70%), and erythromycin (85·48%). Resistance of *S. suis* to many classes of antimicrobial agents such as lincosamides, macrolides, sulphonamides, and TET has already been reported by several authors (Soares *et al.* 2014). There is also evidence of increasing resistance of *S. suis* isolates from both pigs and humans to lincosamides and macrolides in Northern America and European countries (Soares *et al.* 2014; Seitz *et al.* 2016). A high prevalence of TET resistance was reported for *S. suis* isolates in many countries including those of North America, Asia, and some from Europe (Varela *et al.* 2013; Soares *et al.* 2014; Seitz *et al.* 2016). Similarly, high prevalence of TET-resistant *S. suis* isolated from pigs was also found in different regions of China (Huang *et al.* 2015). The resistance to TETs in *S. suis* has become a major worldwide problem, closely related to the widespread use of TET in swine production. Tetracycline resistance has been considered to be an important cofactor in the selection of resistance to macrolides/lincosamides. In this study, 58 of the 62 isolates were resistant to TET, 51 of which were coresistant to macrolides and lincosamides antibiotics.

Antimicrobial resistance patterns of the *S. suis* isolates are shown in Table 2. Sixteen resistance groups (R1 to R16) were observed in this study where 93·54% isolates were multi-drug resistant. It has also been observed that 57 (91·93%) isolates were resistant to at least four antimicrobials. The most predominant resistance pattern observed was CD-COT-E-TE which accounted for 38·70% of the isolates. Although the most prevalent AMR pattern demonstrated by Soares *et al.* (2014) consisted of many (8) antimicrobials, the most prevalent AMR pattern observed in this study was comprised of 4 antimicrobials, which is almost consistent with the findings of other workers (Vela *et al.* 2005; Zhang *et al.* 2008). The percentage of multidrug resistant isolates was found to be lower (93·54%) in this study than those reported by Soares *et al.* (2014) and Soares *et al.* (2015) which were 99·61 and 100%, respectively. The differences observed between countries might be explained by different usage of antimicrobial agents or due to methodological differences between studies.

Streptococcus suis has been the major global driver of antibiotic administration to pigs (Zou *et al.* 2018). Contaminated healthy pigs, especially asymptomatic pigs are clinically important infection sources that increase the risks of potential outbreaks or transmission to clean pigs or humans in close occupational contact with pigs or pork products (Goyette-Desjardins *et al.* 2014). In addition, asymptomatic pigs could potentially serve as reservoirs for multidrug-resistant *S. suis*. Thus, continuous surveillance and monitoring of AMR in *S. suis* isolates is very much essential for effective control of *S. suis* infections in both humans and animals.

To the best of our knowledge, little well-directed data regarding the AMR of *S. suis* isolates especially from clinically healthy pigs have been reported from India. Thus, the prevalence, virulence-associated genes, and AMR patterns among clinically healthy pigs in North East India were investigated in this study. The results could provide a basis for reducing the imprudent use of antimicrobials in pig populations or human cases as well as will throw light on the AMR profiles of *S. suis* in India.

Materials and methods

Sample collection and *S. suis* identification

A total of 304 nasal swab samples were collected from clinically healthy pigs (1–3 months of age) from all North

Table 2 Antimicrobial resistance pattern of *Streptococcus suis* isolates from clinically healthy pigs from North East India

Resistance group	Antimicrobial resistance patterns	Number of isolates
R1	CIP-CEFT	1
R2	CD-CIP	1
R3	CIP-P	2
R4	COT-E-TE	1
R5	CD-COT-E-TE	24
R6	CD-CIP-COT-E-EX-TE	4
R7	CD-CIP-COT-E-HLG-TE	3
R8	CD-CIP-COT-EX-HLS-TE	5
R9	AK-AMP-CD-COT-E-HLG-TE	1
R10	C-CIP-CN-COT-E-EX-TE	1
R11	CD-CIP-COT-E-EX-HLS-TE	8
R12	CD-CIP-COT-E-HLG-HLS-TE	4
R13	CD-COT-E-EX-HLG-HLS-TE	2
R14	AMP-CD-CN-E-EX-HLG-HLS-TE	1
R15	AK-C-CD-CIP-CN-E-EX-HLS-TE	2
R16	AK-C-CD-CIP-COT-E-EX-HLS-TE	2

Eastern states of India during February 2014 to January 2019. Cotton swabs were stored in 1 ml of phosphate-buffered saline and transported to the laboratory at low temperature. The liquid samples were cultured aerobically on Columbia agar plate (Hi-Media, Mumbai, India) supplemented with 5% bovine blood at 37°C and inspected for growth after 24 and 48 h. Colonies 1–2 mm in diameter showing green or brown decolourization around the colony were suspected as potential *S. suis*. The primary identification of the organism was made as per the method of Quinn *et al.* (1994). Further the organism was confirmed as *S. suis* through detection of glutamate dehydrogenase (*gdh*) gene (Okwumabua *et al.* 2003). Representative PCR products were sequenced commercially (Eurofins Genomics India, Bangalore, India), aligned them using MEGA 6.0 (https://www.megasoftware.net), and subjected them to BLAST search (https://blast.ncbi.nlm.nih.gov/Blast.cgi).

Detection of virulence associated genes of *S. suis*

DNA template from each isolate was prepared as per the method described by Okwumabua *et al.* (2003). Amplification of virulence-associated genes such as suilysin (*sly*), muramidase-released protein (*mrp*), and extracellular protein factor (*epf*) including arginine deiminase (*arcA*) was achieved by monoplex PCR assays using the sequence of primers and conditions described previously by Silva *et al.* (2006). Positive and negative control was also used in the study. The positive control used was well characterized *S. suis* isolate (positive for *mrp*, *epf*, and *sly* genes) maintained at the Animal Health Laboratory, National Research Centre on Pig, Indian Council of Agricultural Research, Rani, Guwahati, Assam, India whereas the negative control was nuclease free water.

Antimicrobial susceptibility

Antibiotic susceptibility testing was performed by the disk diffusion method using 14 commercially available antimicrobials disks as per the method of the Clinical and Laboratory Standards Institute guidelines (CLSI 2008). The antimicrobial agents used in this study were: penicillin G (P), 10 IU; ampicillin (AMP), 10 μg; cefalexin (CN), 30 μg; ceftiofur (CEFT), 30 μg; tetracycline (TE), 30 μg; streptomycin (HLS), 300 μg; gentamicin (HLG), 120 μg; amikacin (AK), 30 μg; ciprofloxacin (CIP), 5 μg; enrofloxacin (EX), 10 μg; erythromycin (E), 15 μg; clindamycin (CD), 2 μg; chloramphenicol (C), 30 μg, and co-trimoxazole (COT), 25 μg.

Conflict of Interest

There is no conflict of interest for this manuscript.

References

CLSI/Clinical and Laboratory Standards Institute (2008) *Performance Standards for Antimicrobial Disk and Dilution Susceptibility Tests for Bacteria Isolated from Animals*, 3rd edn. Approved standard M31-A3. Wayne, PA: Clinical and Laboratory Standards Institute.

Espinosa, I., Baez, M., Corona, B., Chong, D., Lobo, E. and Martinez, S. (2013) Molecular typing of *Streptococcus suis* from pigs in Cuba. *Biotecnol Apl* **30**, 39–44.

Goyette-Desjardins, G., Auger, J.P., Xu, J., Segura, M. and Gottschalk, M. (2014) *Streptococcus suis,* an important pig pathogen and emerging zoonotic agent—an update on the worldwide distribution based on serotyping and sequence typing. *Emerg Microbes Infect* **3**, e45.

Gruening, P., Fulde, M., Valentin-Weigand, P. and Goethe, R. (2006) Structure, regulation, and putative function of the arginine deiminase system of *Streptococcus suis*. *J Bacteriol* **188**, 361–369.

Huang, J., Shang, K., Kashif, J. and Wang, L. (2015) Genetic diversity of *Streptococcus suis* isolated from three pig farms of China obtained by acquiring antibiotic resistance genes. *J Sci Food Agr* **95**, 1454–1460.

Ip, M., Fung, K.S.C., Chi, F., Cheuk, E.S.C., Chau, S.S.L., Wong, B.W.H., Lui, S.L., Hui, M. *et al.* (2007) *Streptococcus suis* in Hong Kong. *Diagn Microbiol Infect Dis* **57**, 15–20.

Lacouture, S., Okura, M., Takamatsu, D., Corsaut, L. and Gottschalk, M. (2020) Development of a mismatch amplification mutation assay to correctly serotype isolates of *Streptococcus suis* serotypes 1, 2, 1/2, and 14. *J Vet Diagn Invest* **32**, 490–494.

Lun, Z.R., Wang, Q.P., Chen, X.G., Li, A.X. and Zhu, X.Q. (2007) *Streptococcus suis*: an emerging zoonotic pathogen. *Lancet Infect Dis* **7**, 201–209.

MacInnes, J. and Desrosiers, R. (1999) Agents of the "suicide diseases" of swine: *Actinobacillus suis, Haemophilus parasuis* and *Streptococcus suis*. *Can J Vet Res* **63**, 83–89.

Maneerat, K., Yongkiettrakul, S., Kramomtong, I., Tongtawe, P., Tapchaisri, P., Luangsuk, P., Chaicumpa, W., Gottschalk, M. *et al.* (2013) Virulence genes and genetic diversity of *Streptococcus suis* serotype 2 isolates from Thailand. *Transboun Emerg Dis* **60**, 69–79.

Meekhanon, N., Kaewmongkol, S., Phimpraphai, W., Okura, M., Osaki, M., Sekizaki, T. and Takamatsu, D. (2017) Potentially hazardous *Streptococcus suis* strains latent in asymptomatic pigs in a major swine production area of Thailand. *J Med Microbiol* **66**, 662–669.

Okwumabua, O., Connor, M.O. and Shull, E. (2003) A polymerase chain reaction (PCR) assay specific for *Streptococcus suis* based on the gene encoding the glutamate dehydrogenase. *FEMS Microbiol Lett* **218**, 79–84.

Quinn, P.J., Carter, M.E., Markey, B. and Carter, G.R. (1994) The Streptococci and related cocci. In *Clinical Veterinary*

Microbiology ed. Quinn, P.J. *et al*. pp. 127–135. London: Wolfe Publishing, Mosby-Year Book Europe Limited.

Rui, P., Zhang, Z.Z., Ma, Z.J., Fang, H., Yang, W.J., Zhang, X.Z., Chen, J. and Jia, Q.H. (2012) Detection of virulence-associated factors of *Streptococcus suis* serotype 2 by PCR assay in Hebei, Province of China. *African J Microbiol Res* **6**, 1061–1064.

Schultsz, C., Jansen, E., Keijzers, W., Rothkamp, A., Duim, B., Wagenaar, J.A. and Ende, A. (2012) Differences in the population structure of invasive *Streptococcus suis* strains isolated from pigs and from humans in the Netherlands. *PLoS One* **7**, e33854.

Seitz, M., Valentin-Weigand, P. and Willenborg, J. (2016) Use of antibiotics and antimicrobial resistance in veterinary medicine as exemplified by the swine pathogen *Streptococcus suis*. *Curr Top Microbiol Immunol* **398**, 103–121.

Silva, L.M.G., Baums, C.G., Rehm, T., Wisselink, H.J., Goethe, R. and Weigand, P.V. (2006) Virulence-associated gene profiling of *Streptococcus suis* isolates by PCR. *Vet Microbiol* **115**, 117–127.

Soares, T.C.S., Gottschalk, M., Lacouture, S., Megid, J., Ribolla, P.E.M., Pantoja, J.C.F. and Paes, C.A. (2015) *Streptococcus suis* in employees and the environment of swine slaughterhouses in Sao Paulo, Brazil: occurrence, risk factors, serotype distribution, and antimicrobial susceptibility. *Canadian J Vet Res* **79**, 279–284.

Soares, T.C.S., Paes, A.C., Megid, J., Ribolla, P.E.M., Paduan, K.D.S. and Gottschalk, M. (2014) Antimicrobial susceptibility of *Streptococcus suis* isolated from clinically healthy swine in Brazil. *Canadian J Vet Res* **78**, 145–149.

Suankratay, C., Intalapaporn, P., Nunthapisud, P., Arunyingmongkol, K. and Wilde, H. (2004) *Streptococcus suis* meningitis in Thailand. *Southeast Asian J Trop Med Public Health* **35**, 868–876.

Thi Hoang Mai, N., Thi Hoa, N., Vu Thieu Nga, T., Dieu Linh, L., Thi Hong Chau, T., Xuan Sinh, D., Hoan Phu, N., Van Chuong, L. *et al*. (2008) *Streptococcus suis* meningitis in adults in Vietnam. *Clin Infect Dis* **46**, 659–667.

Varela, N.P., Gadbois, P., Thibault, C., Gottschalk, M., Dick, P. and Wilson, J. (2013) Antimicrobial resistance and prudent drug use for *Streptococcus suis*. *Anim Health Res Rev* **14**, 68–77.

Vela, A.I., Moreno, M.A., Cebolla, J.A., Gonzalez, S., Latre, M.V., Dominguez, L. and Fernandez-Garayzabal, J.F. (2005) Antimicrobial susceptibility of clinical strains of *Streptococcus suis* isolated from pigs in Spain. *Vet Microbiol* **105**, 143–147.

Votsch, D., Willenborg, M., Weldearegay, Y.B. and Weigand, P.V. (2018) *Streptococcus suis*- the “two faces” of a pathobiont in the porcine respiratory tract. *Front Microbiol* **9**, 480.

Wang, K., Zhang, W., Li, X., Lu, C., Chen, J., Fan, W. and Huang, B. (2013) Characterization of *Streptococcus suis* Isolates from Slaughter Swine. *Curr Microbiol* **66**, 344–349.

Winterhoff, N., Goethe, R., Gruening, P., Rohde, M., Kalisz, H., Smith, H.E. and Weigand, P.V. (2002) Identification and characterization of two temperature-induced surface-associated proteins of *Streptococcus suis* with high homologies to members of the arginine deiminase system of *Streptococcus pyogenes*. *J Bacteriol* **184**, 6768–6776.

Xiong, Y., Liu, Q., Qin, F.Y., Bai, Y., Zhu, W., Li, H.M., Guo, J.G., Qin, L. *et al*. (2007) Study on the molecular epidemiology of *Streptococcus suis* type 2 from healthy pigs in Guangx. *Zhonghua Liu Xing Bing Xue Za Zhi* **28**, 593–596.

Zhang, C., Ning, Y., Zhang, Z., Song, L.i., Qiu, H. and Gao, H. (2008) *In vitro* antimicrobial susceptibility of *Streptococcus suis* strains isolated from clinically healthy sows in China. *Vet Microbiol* **131**, 386–392.

Zheng, H., Ji, S., Lan, R., Liu, Z., Bai, X., Zhang, W., Gottschalk, M. and Xua, J. (2014) Population analysis of *Streptococcus suis* isolates from slaughtered swine by use of minimum core genome sequence typing. *J Clin Microbiol* **52**, 3568–3572.

Zou, G., Zhou, J., Xiao, R., Zhang, L., Cheng, Y., Jin, H., Li, L., Zhang, L. *et al*. (2018) Effects of environmental and management-associated factors on prevalence and diversity of *Streptococcus suis* in clinically healthy pig herds in China and the United Kingdom. *Appl Environ Microbiol* **84**, e02590-17.